

Agree, Disagree, Explain: Decomposing Human Label Variation in NLI through the Lens of Explanations

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Abstract

Natural Language Inference (NLI) datasets often exhibit human label variation. To better understand these variations, explanation-based approaches analyze the underlying reasoning behind annotators’ decisions. One such approach is the LiTeX taxonomy, which categorizes free-text explanations in English into reasoning categories. However, previous work applying LiTeX has focused on within-label variation: cases where annotators agree on the NLI label but provide different explanations. This paper broadens the scope by examining how annotators may diverge not only in the reasoning category but also in the labeling. We use explanations as a lens to analyze variation in NLI annotations and to examine individual differences in reasoning. We apply LiTeX to two NLI datasets and align annotation variation from multiple aspects: NLI label agreement, explanation similarity, and taxonomy agreement, with an additional compounding factor of annotators’ selection bias. We observe instances where annotators disagree on the label but provide similar explanations, suggesting that surface-level disagreement may mask underlying agreement in interpretation. Moreover, our analysis reveals individual preferences in explanation strategies and label choices. These findings highlight that agreement in reasoning categories better reflects the semantic similarity of explanations than label agreement alone. Our findings underscore the richness of reasoning-based explanations and the need for caution in treating labels as ground truth.

1 Introduction

Natural Language Inference (NLI) has long served as a benchmark for language understanding (Dagan et al., 2005; Condoravdi et al., 2003), yet annotation divergence, both in labels and in underlying

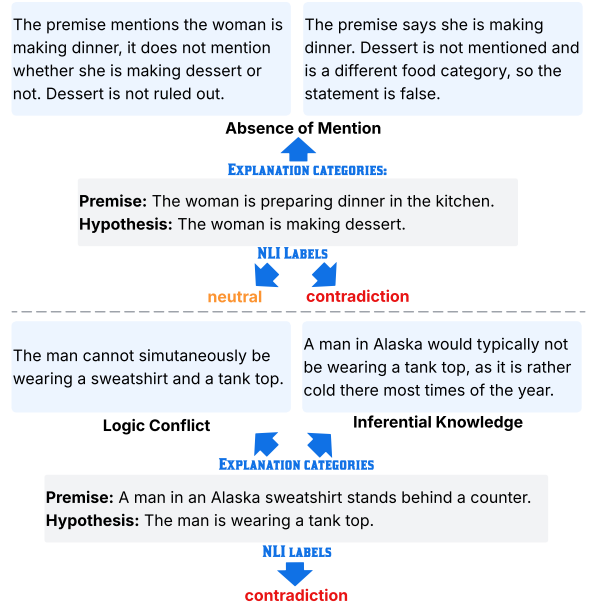


Figure 1: **Agreement and disagreement in NLI annotations through labels and explanations.** The two examples illustrate how annotators may diverge in NLI labels or explanation categories, featuring the LiTeX categories *Logical Conflict* and *Absence of Mention*.

reasoning, has been recognized as a challenge. Recent works have provided new explanations of this phenomenon, drawing on insights from linguistics, pragmatics, and conceptual framing (Plank, 2022; Jiang et al., 2023; Kalouli et al., 2023; Gubelmann et al., 2023).

Explanations written in free-text provide valuable insights into the reasons behind label assignment (Jiang et al., 2023; Tan, 2022; Chen et al., 2025a). Instead of treating NLI labels as isolated outcomes, explanations reveal underlying reasoning strategies that annotators employ. Building on Jiang and de Marneffe (2022), which focuses on categorizing linguistic sources of disagreement in the premise–hypothesis pair, Hong et al. (2025) in-

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roduced the LiTeX taxonomy. The taxonomy is developed based on the e-SNLI dataset [Camburu et al. \(2018\)](#) and categorizes explanations according to eight distinct reasoning strategies.

The LiTeX taxonomy provides a structured framework for analyzing free-text explanations jointly with NLI labels. By considering both label assignments and explanation categories, we can uncover patterns of agreement and disagreement. These distinctions would remain hidden if one only considered the NLI label distribution alone. Incorporating explanation-based reasoning further helps illuminate potential sources of variation in NLI.

Figure 1 illustrates two representative cases of agreement and disagreement when looking into NLI labels and explanations using the LiTeX categories. The top example shows a scenario in which both annotators rely on the *Absence of Mention* rationale, yet assign different labels (neutral vs. contradiction), highlighting divergence in label judgments despite similar explanation strategies. The bottom example shows annotators agreeing on the contradiction label while providing explanations grounded in different categories, reflecting variation in how the relationship between the premise and hypothesis is interpreted.

While LiTeX was originally designed to study within-label variation (like the second example in Figure 1, which provides different explanations for the same NLI label), it remains unclear how well its approach generalizes to label variation settings. In this work, we analyze the interaction of label choices, explanation categories, and explanation text similarities to provide a deeper view of how explanations relate to NLI labels. This perspective allows us to identify patterns of agreement and disagreement that are not apparent from label distribution alone and to better characterize sources of variation across annotators. Specifically, our key contributions include:

From within-label to label variation We annotate two NLI datasets, LiveNLI ([Jiang et al., 2023](#)) and VariErr ([Weber-Genzel et al., 2024](#)) with the LiTeX taxonomy, enabling the study of reasoning categories across both within-label and label variation.

Annotator tracking via explanation categories

By combining NLI labels with explanation categories, we track individual annotators’ reasoning patterns and uncover behavioral consistencies that are not apparent from label distributions alone.

Quantitative analysis of agreement beyond labels

We measure agreement at three levels: NLI label, explanation category, and text similarity of explanations. Our results show that alignment in reasoning categories better correlates with explanation similarity than NLI label agreement alone, emphasizing the importance of explanations for understanding annotator disagreement.

2 Related Work

Most benchmark NLI datasets provide multiple annotations per instance, enabling the study of annotation variation. For example, the Stanford NLI (SNLI; [Bowman et al. 2015](#)) and MultiNLI ([Williams et al., 2018](#)) datasets collect multiple crowd-sourced judgments for each premise–hypothesis pair, motivating work on systematic disagreement and label variation, often through re-annotation and analyses specifically examining disagreement ([Pavlick and Kwiatkowski, 2019](#); [Kalouli et al., 2023](#)). To further address this, adversarial datasets such as ANLI ([Nie et al., 2020a](#)) and resources targeting different aspects of annotation variation, such as ChaosNLI ([Nie et al., 2020b](#)) for annotator disagreement and AmbiEnt ([Liu et al., 2023](#)) for ambiguity, are introduced.

Closer to our line of work, recent datasets use free-text explanations and highlights to reveal the reasoning behind NLI labels ([Camburu et al., 2018](#); [Jiang et al., 2023](#); [Nigohjkar et al., 2023](#); [Weber-Genzel et al., 2024](#)). e-SNLI expanded SNLI ([Bowman et al., 2015](#)) by crowd-sourcing highlight and explanation annotations on the pre-annotated majority label. LiveNLI ([Jiang et al., 2023](#)) recruited crowd-workers to annotate NLI labels while also providing highlights and explanations (i.e., ecologically valid explanations, produced jointly with the label rather than post hoc). VariErr ([Weber-Genzel et al., 2024](#)) use such ecologically valid explanations as a foundation for error detection. [Hong et al. \(2025\)](#) built a taxonomy, LiTeX, to categorize these free-text explanations, but its scope was limited to e-SNLI and focused on analyzing within-label variation.

A parallel line of work has examined *annotator disagreement*. For example, [de Marneffe et al. \(2012\)](#) and [Uma et al. \(2022\)](#) identified structured patterns of disagreement. For NLI, understanding human reasoning is crucial to interpreting agreement or disagreement. Annotators often rely on various reasoning strategies, such as substitution, nega-

tion, bridging inferences, and world knowledge inference (Jiang and de Marneffe, 2022; Kalouli et al., 2023; Sanyal et al., 2024; Hong et al., 2025).

Prior work on annotator decisions often focuses on the relationship between the premise-hypothesis pair and the resulting NLI label, but the correlation with explanations remains underexplored. In particular, it is unclear how similar explanation categories can lead to different labels, or how the same label may reflect distinct rationales. We address this gap by using the LiTEX taxonomy to analyze the interaction between explanation categories and label assignments, providing a more fine-grained view of variations in NLI annotations.

3 From Within-label Variation to Label Variation

To characterize patterns of variation in different NLI datasets, we use the LiTEX taxonomy (Hong et al., 2025) and apply it to two additional English NLI datasets exhibiting label variation. We then examine the co-occurrence patterns between NLI labels and explanation categories across datasets.

3.1 LiTEX: a Linguistic Taxonomy of Explanations

LiTEX categorizes NLI explanations into two reasoning categories: *Text-Based (TB)* and *World-Knowledge (WK)*. TB draws on linguistic evidence in the premise and hypothesis and comprises six categories: *Coreference*, *Syntactic*, *Semantic*, *Pragmatic*, *Absence of Mention*, and *Logic Conflict*. WK invokes background knowledge beyond text, covering *Factual Knowledge* and *Inferential Knowledge*. For detailed definitions and annotated examples of all categories, please refer to Hong et al. (2025).

LiTEX was originally developed on free-text explanations in e-SNLI (Camburu et al., 2018) to characterize within-label variation, where annotators reach the same label via different rationales (Jiang et al., 2023; Hong et al., 2025). In this paper, we use LiTEX beyond its original scope by applying it to additional datasets, examining reasoning strategies not only within-label but also across labels. Furthermore, we use the taxonomy to analyze annotator labeling behavior and to deepen our understanding of the relationship between explanations and NLI labels.

3.2 Annotation on LiveNLI and VariErr

To study how annotators diverge in both reasoning and label selection, we apply LiTEX to two En-

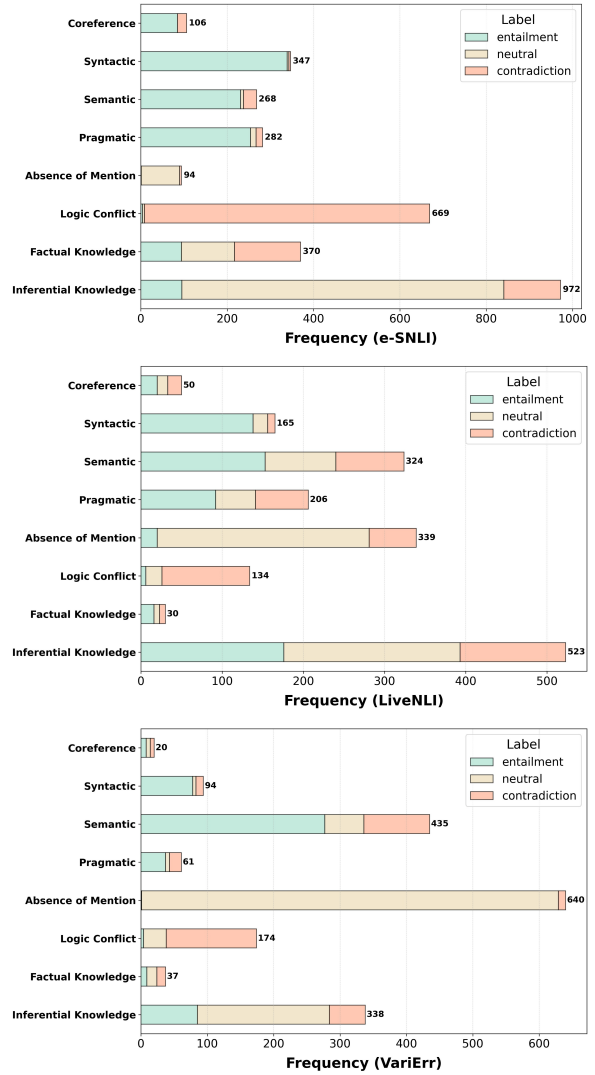


Figure 2: Co-occurrence of LiTEX explanation categories and NLI labels across the e-SNLI, LiveNLI, and VariErr datasets.

glish datasets with label variation and explanation annotations — **LiveNLI** (Jiang et al., 2023) and **VariErr** (Weber-Genzel et al., 2024).

LiveNLI is a high-quality explanation dataset derived from a subset of MNLi (Williams et al., 2018), containing 122 NLI items. Each item is annotated by at least 10 crowdworkers, who assign one or more NLI labels (*true*, *either*, *false*), highlight relevant spans, and provide free-text explanations (Jiang et al., 2023). The dataset contains 1,404 explanation-label pairs. For the consistency of our analyses and to align with e-SNLI and VariErr, we map (*true*, *either*, *false*) to (*entailment*, *neutral*, *contradiction*).

VariErr complements LiveNLI by focusing on variation and errors in English NLI. It consists of 1,933 model-generated explanations for 500 re-

annotated MNLI items, along with 7,732 human validity judgments (Weber-Genzel et al., 2024). LiveNLI captures natural annotator disagreement, and VariErr introduces both plausible alternative explanations and annotation errors.

A key difference from e-SNLI lies in the labeling scheme. While e-SNLI assigns a single gold label, both LiveNLI and VariErr allow multiple plausible labels per instance and are ecologically valid, with labels and explanations annotated by the same people. These features allow us to examine the relationship between label assignments and explanation categories. Appendix A presents a detailed analysis of label distribution per NLI item in the two datasets.

We use LiTeX to annotate all explanations from LiveNLI and VariErr. All annotations are carried out by a trained annotator. The annotator is instructed to categorize free-text explanations according to the reasoning explicitly expressed within them and is asked to select the most prominent explanation categories from LiTeX. This design follows the original setup in which LiTeX was introduced, ensuring alignment with prior work. To measure inter-annotator agreement (IAA), we recruited a second annotator to annotate 100 explanations from each dataset independently.¹ We obtained a Cohen’s Kappa (κ) of 0.828 for LiveNLI and 0.792 for VariErr, similar to the IAA on e-SNLI (Hong et al., 2025). For a more detailed analysis of IAA results and per-category agreement, please refer to Appendix B.

3.3 LiTeX Categories across NLI Labels

Figure 2 shows the distributions of the LiTeX categories across NLI labels in three datasets: e-SNLI, LiveNLI and VariErr. In terms of frequency of the LiTeX categories, we see that *Coreference* appears less frequently in all datasets compared to the other categories. Notable differences can also be observed across datasets. For example, *Inferential Knowledge* is the dominant category in both e-SNLI and LiveNLI but is less prominent in VariErr. In contrast, *Absence of Mention* is the most frequent category in VariErr, ranks second in LiveNLI, and occurs relatively less often in e-SNLI.

Additional interesting patterns arise in the **co-occurrences** between the distributions of NLI labels within taxonomy categories. The *neutral* la-

bel dominates the *Absence of Mention* category across all three datasets. This is consistent with the nature of this reasoning category, which focuses on information gaps between the premise and hypothesis. For *Factual Knowledge* and *Inferential Knowledge*, explanations are distributed relatively evenly across the three NLI labels. This reflects the fact that these categories involve the introduction of external or inferred knowledge, without a strong bias toward a specific label. The *Syntactic*, *Semantic*, and *Pragmatic* categories are more strongly associated with entailment (despite slight variations across datasets), suggesting that annotators often rely on evidence from different linguistic levels within the premise and hypothesis to establish entailment relationships.

In sum, although the absolute distributions of explanation categories differ across datasets, the label distribution and dominant label for each category remain highly consistent. The observed differences in category distribution may stem from factors such as annotator backgrounds and preferences, dataset-specific item selection, and annotation guidelines. Nevertheless, the stable category-label co-occurrence patterns indicate that, despite being originally developed only on the e-SNLI dataset, LiTeX provides **a reliable characterization of reasoning categories that generalize across explanations** in different NLI datasets.

4 Label and Reasoning Preferences among Individual Annotators

To better understand disagreement in NLI, we analyze annotator preferences over NLI labels and explanation categories. We focus on four annotators in LiveNLI and VariErr, as e-SNLI does not provide annotator identifiers for tracking. Specifically, we track individual annotators across two dimensions: their **NLI label preferences** (e.g., tendency to overuse “neutral”), and their **reasoning-category preferences**, taxonomy classification of their free-text explanations annotated in this paper.

Unlike VariErr, in which all items are annotated by the same four annotators, LiveNLI involves a much larger pool of annotators (Jiang et al., 2023). The number of items annotated by the same annotator ranges from 1 to 122. To align with the setup in VariErr, we chose a group of four annotators {w1, w4, w7, w8} from LiveNLI who have the highest number of overlapping annotated items (115 in total). Appendix C provides detailed tables of

¹Both annotators were students trained with the taxonomy definitions and examples, and were paid according to the national standard.

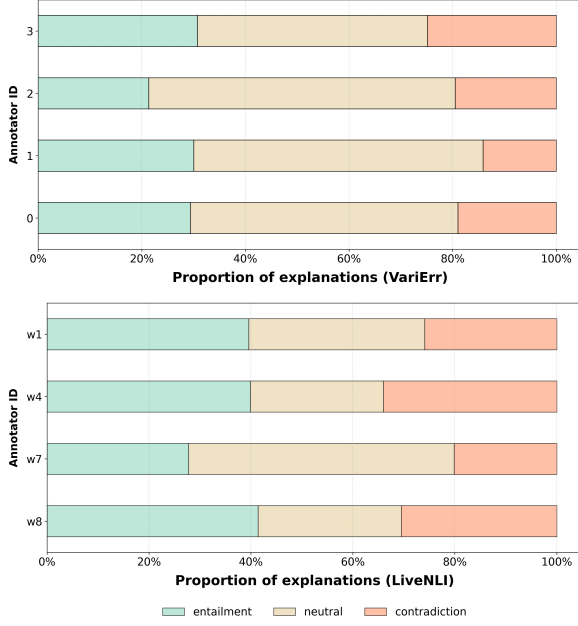


Figure 3: Distribution of NLI labels (entailment, neutral, contradiction) across LiveNLI and VariErr annotators. The legend at the bottom specifies the color-label correspondence, while the area of each color segment represents the number of instances assigned to that label.

NLI label distributions and explanation category preferences for the four annotators in LiveNLI and the four annotators of VariErr, computed over the intersection of items they annotated.

Label Preferences Figure 3 illustrates the distribution of NLI labels assigned by four annotators for VariErr and for LiveNLI. For VariErr, while all annotators exhibit a noticeable preference for the *neutral* label, the degree of this preference varies. Annotators 0, 1, and 2 all assign *neutral* in over 50% of their examples, with Annotator 2 showing the strongest bias—nearly 60% of their annotations are *neutral*. In contrast, Annotator 3 demonstrates a more balanced labeling behavior, with a more even spread across *entailment*, *neutral*, and *contradiction*, resulting in the lowest *neutral* proportion (44.42%).

As for LiveNLI, three annotators (w1, w4, w8) show a slight preference for the *entailment* label, each assigning it to approximately 40% of the items. In contrast, annotator w7 exhibits a stronger preference for the *neutral* label, assigning it in over half of the cases (52.08%). The *contradiction* labels are the least common overall. Compared to annotators in VariErr, where *neutral* was the dominant label for all four annotators, the LiveNLI group shows more varied labeling tendencies.

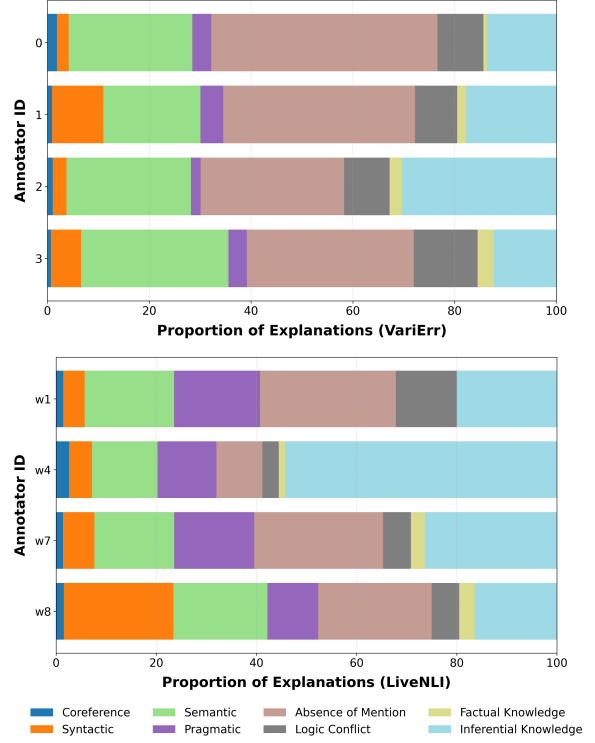


Figure 4: Distribution of explanation category per annotators in LiveNLI and VariErr. Colors correspond to different explanation categories.

Reasoning-Type Preferences Figure 4 presents the distribution of explanation categories used by the annotators. Individual differences emerge in how annotators ground their inferences. For VariErr, annotator 0 shows a dominant reliance on *Absence of Mention* (44.44%) and *Semantic* reasoning (24.31%), with only minimal use of world knowledge-based categories such as *Factual Knowledge* (0.69%) and *Inferential Knowledge* (13.66%). This pattern suggests a preference for surface-level paraphrastic inference, rather than deeper reasoning. In contrast, Annotator 1 exhibits a more balanced distribution, with moderate use of the reasoning strategies. Annotator 2 stands out with a very strong emphasis on *Inferential Knowledge* (30.29%), while still relying on *Semantic* explanations (24.48%). This suggests a knowledge-intensive reasoning, grounded in world knowledge and inferencing beyond what is stated. Similarly, Annotator 3 relies more on knowledge-based reasoning, using *Inferential Knowledge* (12.3%) and *Factual Knowledge* (3.2%) more frequently.

For LiveNLI, several trends emerge from the distribution. First, *Semantic* and *Absence of Mention* explanations are consistently among the most frequently used categories across annotators, sug-

Agreement Class	Entropy	Support (%)	Category Agreement	token 1-gram	token 2-gram	POS 1-gram	POS 2-gram	cosine (%)	euclidean (%)
<i>VariErr</i>									
Full (4-0-0)	0.00	43.75	0.76	35.05	11.53	74.21	35.06	52.87	51.89
Partial (3-1-0)	0.81	28.95	0.60	34.72	10.62	78.31	34.85	52.81	51.96
Two Pairs (2-2-0)	1.00	23.36	0.56	30.80	8.50	73.47	31.23	49.22	51.02
Divergent (2-1-1)	1.50	3.95	0.50	32.02	9.96	70.37	31.50	48.21	50.91
<i>LiveNLI</i>									
Full (4-0-0)	0.00	21.74	0.62	40.31	10.26	88.89	41.96	58.05	53.24
Partial (3-1-0)	0.81	34.78	0.56	40.44	11.95	86.99	44.02	54.47	52.27
Two Pairs (2-2-0)	1.00	23.48	0.60	38.97	10.67	88.38	43.24	55.33	52.84
Divergent (2-1-1)	1.50	20.00	0.54	36.61	8.99	85.09	41.44	53.63	52.35

Table 1: Aggregated statistics across agreement classes for LiveNLI and VariErr, based on how many annotators agree on the NLI label. We report the label entropy, the percentages of support items, corresponding category agreement, and average pairwise explanation similarity. Color coding highlights relative deviations within each dataset: cells shaded in blue or red indicate values that are higher or lower than expected given the level of label agreement. Darker shades correspond to larger deviations in ranking, and lighter shades indicate smaller deviations.

gesting that both lexical-semantic inferences and missing information play a central role. Second, we observe notable variation in the use of *Inferential Knowledge*: annotator w4 relies on this category in over half of their explanations (54.25%), while w1 and w8 use it far less frequently (20.00% and 16.41% respectively), indicating divergent preferences in relying on external world knowledge. Similarly, *Syntactic* explanations are prominent for w8 (21.88%) compared to the others, reflecting a possible inclination toward structural reasoning. Conversely, *Logical Conflict*, *Factual Knowledge*, and *Coreference* are relatively rare across annotators, suggesting these reasoning categories are less frequently invoked or salient in this LiveNLI subset.

Overall, the annotator-level analysis facilitated by LiTeX over the two datasets reveals that **different annotators tend to adopt distinct reasoning strategies**—arriving at different NLI labels for the same premise–hypothesis pairs. Observing only the distribution of NLI labels is insufficient to uncover the underlying reasoning rationales. To gain a deeper understanding of annotator behavior, we next conduct a fine-grained item-level analysis that disentangles variation in reasoning from variation in label assignment.

5 Measuring and Interpreting Agreement

Building on observations in §4 that NLI label distributions are insufficient to uncover reasoning rationales of individual annotators, this section takes a closer look at how to measure and interpret agreement and disagreement in NLI tasks *at the instance level*. We first quantify annotator agreement across

three dimensions: NLI labels, explanation categories, and textual similarity between explanations. We then compute pairwise agreement and visualize it using conditional probability heatmaps to examine the relationship between explanation categories and labeling. Finally, we present an example that is covered in both datasets to illustrate how annotators may align or diverge in their label decisions and explanations.

5.1 Quantifying Annotator Agreement Beyond Labels

To gain a clearer picture of annotator agreement on NLI labels and LiTeX categories, we group NLI items based on the set of annotators who labeled them and how often these annotators agreed with each other. We define four NLI **label agreement** classes: *Full Agreement* (4-0-0) indicates all four annotators assigned the same label; *Partial Agreement* (3-1-0) refers to cases where three annotators agreed while one differed; *Two Pairs Agreement* (2-2-0) denotes a balanced split, with two annotators agreeing on one label and the other two on a different label; and *Divergent* (2-1-1) captures maximal disagreement, where three different labels are assigned.

Category agreement is measured via the average Jaccard similarity between the explanation categories. Since each explanation has only one category, this reduces to computing the proportion of explanation pairs that share the same category:

$$\text{Jaccard}(a, b) = \begin{cases} 1, & \text{if } a = b \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

For example, for an item with four annotators ($\binom{4}{2} = 6$ pairs), the LiveNLI example in Table 2 shows a 2–2 split across categories (2/6 matches), while the VariErr example shows a 3–1 split (3/6 matches), corresponding to agreement scores of 0.33 and 0.5.

We follow Chen et al. (2025b) and Hong et al. (2025) to quantify **textual similarities between explanations** using measures from Giu-lianelli et al. (2023). Lexical and syntactic similarities evaluate overlapping unigrams and bigrams on tokens and POS tags. We use cosine and Euclidean to measure semantic similarity between sentence embeddings, obtained using the all-distilroberta-v1² model from Sentence-Transformers (Sanh et al., 2019). Scores are averaged pairwise across four explanations.

Table 1 summarizes the label entropy, number of supporting items, category agreement, and average explanation similarity for each label agreement class. Examining across evaluation metrics, the color coding highlights deviations in ranking across agreement classes relative to label agreement, though the absolute differences remain small.

VariErr generally exhibits less ranking deviation than LiveNLI, with a matched ranking between label agreement class and category agreement, and only light deviations in textual similarity measures. Full agreement is also considerably more frequent in VariErr (43.75%) than in LiveNLI (21.74%), indicating that annotators tend to reach agreement more easily in VariErr. Moreover, all ranking deviations in VariErr concern an additional NLI label: between full and partial (4-0-0 vs. 3-1-0) and between two pairs and divergent (2-2-0 vs. 2-1-1), whereas many LiveNLI ranking deviations stem from different distributions of the same labels, i.e., between partial and two pairs (3-1-0 vs. 2-2-0).

Looking at the text similarity measures, we found that cosine similarity resonates the most with label agreement on both datasets, exhibiting moderate differences in scores across classes and only minor deviations in ranking on LiveNLI between partial and two pairs. More interestingly, the pattern of cosine similarity aligns more closely with category agreement than label agreement. This observation **tentatively suggests that shared reasoning categories may better capture the semantic similarity of explanations than label agreement.**

²<https://huggingface.co/sentence-transformers/all-distilroberta-v1>

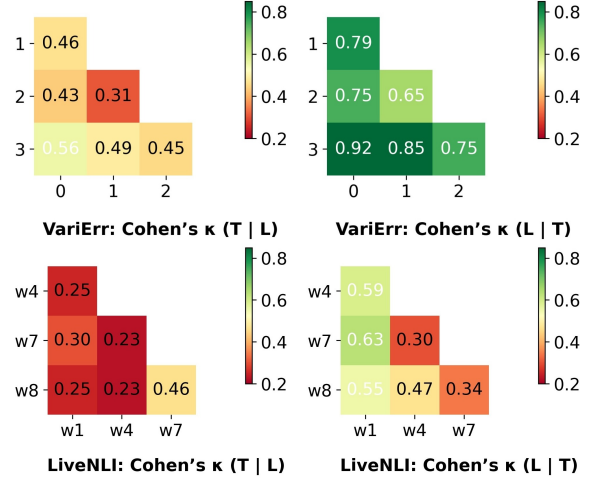


Figure 5: Pairwise annotator agreement (conditional Cohen’s κ) between taxonomy matches (T) and label matches (L).

5.2 Pairwise Annotator Agreement on Reasoning Category and Labeling

To quantify how annotators align explanation strategies with label assignments, we examine pairwise agreement patterns between explanation categories (reasoning) and NLI labels (labeling) using conditional Cohen’s κ scores (Figure 5). This analysis focuses on the interaction between explanation categories and NLI labels, allowing us to examine variation across both. Specifically, for each pair of annotators, we compute two conditional κ scores:

$$\kappa(T | L), \quad (2)$$

which measures taxonomy category agreement (reasoning alignment) on the subset of instances where the two annotators agree on the NLI label, and

$$\kappa(L | T). \quad (3)$$

which measures label agreement on the subset of instances where the two annotators match in their reasoning categories. Unlike raw agreement, Cohen’s κ adjusts for chance, providing a more robust measure of inter-annotator alignment.

VariErr shows higher agreement than LiveNLI under both conditional κ metric, indicating more stable reasoning–label mappings in VariErr, whereas LiveNLI reflects greater variation in annotator interpretations. Across both datasets, annotators show higher κ for labels conditioned on taxonomy category matches $\kappa(L | T)$ than for taxonomy category conditioned on label matches $\kappa(T | L)$. Our approach can explain the asymmetry between

Premise: The author began with a set of hunches or hypotheses about what can go wrong in agency management, and what would be evidence supporting—or contradicting—these hypotheses.
Hypothesis: The hunches provided by the author weren’t realistic as it pertains to agency management.
[CHAOSNLI] [E,N,C]: [0.64, 0.06, 0.30]

Data	Ann.	NLI Label	Explanation	Exp. Category
LiveNLI	w1	Neutral	The context notes that the hunches were provided, but there is no information on their veracity or plausibility. Thus, the statement could be true or false, as it is not known whether they were realistic based on the provided information.	Absence of Mention
	w4	Contradiction	If an author is planning to write about a certain topic, they likely have enough knowledge to form informed opinions. Thus, it is much more likely that the statement is false, since these opinions would be at least somewhat realistic.	Inferential Knowledge
	w7	Contradiction	The author gave evidence to support the hunches, it is unlikely that the hunches were unrealistic.	Inferential Knowledge
	w8	Contradiction	It was not stated that the hunches were unrealistic.	Absence of Mention
VariErr	0	Neutral	It is not clear whether the hunches provided by the author were realistic or not.	Absence of Mention
	1	Neutral	It is not clear how realistic the hypotheses were.	Absence of Mention
	2	Neutral	The judgment of the hunches is not given in the context.	Absence of Mention
	3	Contradiction	The hunches could be realistic, as author provides potential evidence supporting these hypotheses.	Inferential Knowledge

Table 2: An NLI item annotated in both LiveNLI and VariErr. Despite identical inputs, annotators diverge in NLI labels and explanation strategies.

the two conditional probabilities. When annotators share the same taxonomy categories, they are highly likely to assign the same label, indicating that divergence in NLI label assignments exists but occurs less frequently. In contrast, when annotators agree on the final label, their reasoning categories often diverge, suggesting that **divergence in the explanation reasoning categories is relatively more common than that in the labeling**. Therefore, to capture label variation more accurately and informatively, it is crucial to focus on the reasoning explicitly expressed through free-text explanations, as it is the dominant source of variation.

5.3 Example: Diverging Interpretations on the Same NLI Instance

To conclude our analysis, we present an example that illustrates how annotators from two different datasets interpret the same NLI instance in divergent ways, both in the reasons exhibited in their explanations and in the label they choose. Table 2 shows one item annotated by the eight annotators we analyzed in the earlier sections, four from LiveNLI and four from VariErr. For additional qualitative illustrations, further examples are provided in the Appendix D.

Comparing the two datasets, we observe that

while the sets of chosen NLI labels (neutral and contradiction) and explanation categories (*Absence of Mention* and *Inferential Knowledge*) are the same, the distribution of these choices differs: in LiveNLI, three annotators opted for contradiction, whereas in VariErr, three chose neutral. In terms of explanation categories, LiveNLI annotators are evenly split, with two selecting *Absence of Mention* and two choosing *Inferential Knowledge*. In VariErr, three annotators attribute their reasoning to *Absence of Mention*, while one goes for *Inferential Knowledge*.

The example further illustrates patterns of agreement and disagreement in NLI annotations. Among LiveNLI annotators, w4, w7, and w8 agree on the NLI label (contradiction), but provide explanations grounded in different reasoning categories, namely *Inferential Knowledge* vs. *Absence of Mention*, showcasing within-label variation and divergence at reasoning. In contrast, w1 and w8 both provide explanations categorized as *Absence of Mention*, yet arrive at different NLI labels (neutral vs. contradiction), pointing to divergence at the labeling. Meanwhile, VariErr annotators demonstrate both label and reasoning agreement. Annotators 0, 1, and 2 all classify the instance as neutral, supported by nearly identical explanations and shared cate-

gorization as *Absence of Mention*. This coherence suggests a degree of alignment between reasoning categories and NLI labels, from interpretive rationale to label decision. Annotator 3 stands out with Inferential Knowledge reasoning and a contradiction label, further emphasizing how divergent reasoning can lead to different label choices.

Overall, this example shows **how combining NLI labels with explanation categories reveals deeper patterns of disagreement and agreement**—distinctions that would remain hidden if one only considered label distributions alone, while reasoning information further illuminates the underlying sources of variation.

6 Conclusion

Understanding why annotators diverge is key to interpreting NLI labels. We extend LiTeX to two NLI datasets with free-text explanations and jointly analyze within-label and cross-annotator label variation. By combining labels, taxonomy categories, and explanation texts, we uncover reasoning patterns that label distributions do not capture. Across analyses, taxonomy categories track explanation-text similarity more closely than labels, emphasizing reasoning paths over surface label agreement.

Our results suggest three broader implications. First, NLI explanations differ in what they ground the decision on, and these differences correlate with how often annotators agree or disagree. Second, agreement in reasoning type is more predictive of label agreement than shared labels are of reasoning-type agreement, indicating that reasoning provides a key trace of variation. Third, annotators exhibit distinct reasoning profiles, which can lead to divergent labels on the same instances. These findings motivate future NLI work to explicitly track reasoning categories during dataset construction (e.g., which categories appear and which ones tend to trigger disagreement) and to use reasoning categories to stratify evaluation sets, offering a more diagnostic view of where models succeed or fail.

More broadly, labels and even aggregated label distributions can hide how annotators arrive at their interpretations. Complementing labels with explanations provides a clearer window into patterns of agreement and disagreement among annotators. While the ideal explanation format remains open, free-text explanations retain a unique advantage in revealing how different interpretations can emerge from the same input.

Future work can extend LiTeX to allow multiple reasoning strategies per explanation, capturing more interactions. Modeling annotator backgrounds may further reveal systematic sources of disagreement. Finally, integrating this framework with explanation generation could improve the quality and evaluation of model rationales, and applying it to other domains would test its generality. Annotations and analysis are publicly available at <https://github.com/mainlp/LiTeX-NLI-extension>.

Limitations

Our work has several limitations. First, our analysis relies on the category set defined in LiTeX, which may not fully capture the complexity or compositionality of human reasoning; extending the taxonomy to allow multiple categories per explanation is an important direction.

Second, our datasets are substantially smaller than large-scale NLI benchmarks such as SNLI or MNLI. However, the analysis is based on 1,404 explanations from LiveNLI and 1,933 explanations from VariErr, which is relatively large for manually categorized free-text explanations. Accordingly, our goal is not to estimate fine-grained population statistics for all possible annotators, but to uncover robust qualitative and quantitative patterns in how reasoning categories relate to label variation.

Third, our annotator-level analysis focuses on annotators with sufficient per-annotator coverage: we analyze four annotators who each contributed explanations for over 100 items on a shared set of instances, providing within-annotator evidence for comparing reasoning profiles. Nevertheless, including more annotators and more diverse annotator populations would strengthen our work.

Finally, we do not directly model annotator-pool effects (e.g., background knowledge, fatigue, or instruction framing), and our similarity measures (e.g., sentence-embedding cosine similarity and Jaccard agreement over categories) provide only a partial view of reasoning variation, potentially missing deeper pragmatic or structural divergences.

Ethical considerations

We do not foresee any ethical concerns associated with this work. All analyses were conducted using publicly available datasets and models. No private or sensitive information was used.

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A Label Distribution per NLI Item in LiveNLI and VariErr

To better understand the distribution characteristics of NLI labels in the selected two datasets, we visualize the aggregated label probabilities for each item.

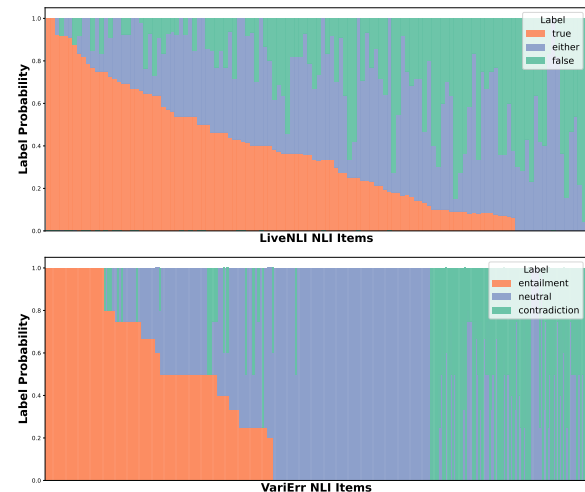


Figure 6: Normalized label distribution per NLI item in LiveNLI and VariErr. Items are sorted by the proportion of *entailment/true* labels.

Figure 6 presents stacked bar charts of the normalized NLI label distributions across all items, sorted by the proportion of *entailment/true* labels to highlight overall patterns of annotator agreement. Both datasets reveal a range of label variation, with many items reflecting ambiguity or disagreement among annotators. However, their distributions differ in characteristic ways. LiveNLI shows greater diversity, particularly in cases where the *either* label dominates or co-occurs substantially with the other two categories. In contrast, VariErr exhibits more concentrated distributions, with fewer items

displaying high levels of ambiguity. Nonetheless, some items still reveal substantial variation in label assignment, pointing to challenging or underspecified NLI cases. These findings motivate our subsequent analysis of how explanation similarity varies across explanation categories and NLI labels.

B Inter-annotation agreement (IAA) results of LiveNLI and VariErr

To further strengthen the validity and transparency of our annotation process, we report detailed IAA statistics for both **LiveNLI** and **VariErr**.

As shown in Table 3, we observe that the distribution of explanation categories is notably non-uniform across both datasets. Certain categories, such as *Factual Knowledge*, occur only rarely compared to more dominant categories like *Absence of Mention* or *Pragmatic*. This skewed distribution is consistent with observations reported in the original LITEX framework (Hong et al., 2025). Given this imbalance, our analysis does not aim to draw fine-grained conclusions about every individual explanation category.

Category	LiveNLI		VariErr	
	Annotator 1	Annotator 2	Annotator 1	Annotator 2
Absence of Mention	39	41	36	38
Pragmatic	21	14	19	17
Inferential Knowledge	18	14	16	15
Semantic	6	16	10	12
Logic Conflict	6	5	7	6
Syntactic	5	4	5	4
Factual Knowledge	3	4	4	5
Coreference	2	2	3	3
Total	100	100	100	100

Table 3: Distribution of LITEX categories in the 100-item IAA samples for LiveNLI and VariErr.

Table 4 reports per-category agreement across the two datasets. We observe systematic variation in IAA: high-frequency categories such as *Absence of Mention* show consistently high agreement (90% and 88%), indicating that they are salient and reliably identifiable. Categories like *Inferential Knowledge* and *Logic Conflict* also achieve relatively strong agreement (around 80%), reflecting well-defined reasoning patterns. In contrast, *Factual Knowledge* shows lower agreement (67% and 70%), likely due to limited sample size and overlap with categories such as *Inferential Knowledge*.

C Annotator-wise distribution of NLI labels and explanation categories

Table 5 presents the detailed NLI label distributions and explanation category preferences for the four

Category	LiveNLI		VariErr	
	Support	Agree (%)	Support	Agree (%)
Absence of Mention	40	90	38	88
Pragmatic	17	76	18	78
Inferential Knowledge	16	81	15	80
Semantic	11	73	12	75
Logic Conflict	6	83	7	82
Syntactic	4	75	5	74
Factual Knowledge	3	67	4	70
Coreference	2	100	3	100

Table 4: Per-category raw agreement for LiveNLI and VariErr.

annotators in LiveNLI and VariErr, computed over the intersection of items they jointly annotated in each dataset. This table provides the full statistics underlying the summary reported in Section 4.

D NLI items annotated in both LiveNLI and VariErr

Table 6 presents more illustrative examples of NLI items that were annotated in both LiveNLI and VariErr by the selected annotator groups. These examples provide additional qualitative context for the agreement patterns discussed in the main text.

Annotator ID	#Ex	Ent	Neu	Con	Coref	Synt	Sem	Prag	Abs	Logic	FK	IK
<i>VariErr</i>												
0	432	29.40	51.62	18.98	1.85	2.31	24.31	3.70	44.44	9.03	0.69	13.66
1	446	30.04	55.83	14.13	0.90	10.09	19.06	4.48	37.67	8.30	1.79	17.71
2	482	21.37	59.13	19.50	1.04	2.70	24.48	1.87	28.22	8.92	2.49	30.29
3	439	30.75	44.42	24.83	0.68	5.92	28.93	3.64	32.80	12.53	3.19	12.30
<i>LiveNLI</i>												
w1	140	39.29	34.29	25.71	1.43	4.29	17.86	17.14	27.14	12.14	0.00	20.00
w4	153	39.87	26.14	33.99	2.61	4.58	13.07	11.76	9.15	3.27	1.31	54.25
w7	144	27.78	52.08	20.14	1.39	6.25	15.97	15.97	25.69	5.56	2.78	26.39
w8	128	41.41	28.13	30.47	1.56	21.88	18.75	10.16	22.66	5.47	3.13	16.41

Table 5: Annotator-wise distribution of NLI labels (entailment, neutral, contradiction) and explanation categories in LiveNLI and VariErr. Percentages (%) are shown for both types of distributions.

Premise: If you have any questions about this report, please contact Henry R. Wray, Senior Associate General Counsel, at (202) 512-8581. Hypothesis: Henry R. Wray can be reached at (555) 512-8581. [CHAOSNLI] [E,N,C]: [0.21, 0.76, 0.03]				
Dataset	Annotator	NLI Label	Explanation	Explanation Category
LiveNLI	w1	Contradiction	The two phone numbers provided for Wray do not match, thus the statement is likely to be false since it does not present the same number as in the context.	Logic Conflict
	w4	Contradiction	Henry R. Wray's actual phone number is listed in the context. While it's true he could have multiple contact numbers, it's more likely that the context is his only contact number since it's connected to his business.	Inferential Knowledge
	w7	Contradiction	The phone number for Henry Wray is (202) 512-8581, so (555) 512-8581 is probably not going to reach him.	Inferential Knowledge
	w8	Contradiction	Henry R. Wray can be reached at a difference number which is (202) 512-8581.	Semantic
VariErr	0	Contradiction	The number of Henry R. Wray mentioned in the statement is wrong.	Logic Conflict
	1	Contradiction	The phone number starts with (202) not with (555).	Logic Conflict
	2	Contradiction	The number is wrong, it should be (202) 512-8581 not (555) 512-8581.	Logic Conflict
	3	Contradiction	wrong phone numbers.	Logic Conflict
Premise: It is, as you see, highly magnified. Hypothesis: It is plain for you to see that it is amplified. [CHAOSNLI] [E,N,C]: [0.76, 0.10, 0.14]				
LiveNLI	w1	Entailment	Magnification is a form of amplification. Therefore, if something is easily seen as magnified, it follows that it's plain to see it's amplified.	Inferential Knowledge
	w4	Entailment	Whatever the speaker is pointing out is clearly magnified or "amplified", and the person they are showing this to is merely having this pointed out to them, making the statement most likely true.	Inferential Knowledge
	w7	Entailment	The words magnified and amplified have the same basic meaning, so the statement agrees with the context.	Semantic
	w8	Entailment	Both amplified and magnified mean the same thing so the statement is true.	Semantic
VariErr	0	Entailment	Both the context and the statement suggest that it is magnified.	Semantic
	1	Entailment	The statement is a paraphrase of the context.	Syntactic
	2	Entailment	highly magnified can be interpreted amplified.	Semantic
	3	Entailment	It can be seen, and it is magnified.	Semantic
Premise: A clean, wholesome-looking woman opened it. Hypothesis: The woman was trying to be desecrate. [CHAOSNLI] [E,N,C]: [0.68, 0.31, 0.01]				
LiveNLI	w1	Contradiction	The context notes that the woman is clean and wholesome-looking while the statement notes that the woman was being disrespectful, which is not compatible. Thus, it is likely to be false.	Semantic
	w4	Neutral	The statement is nonsensical. Hence there's no information in it, either true or false, to be compared to the context.	Absence of Mention
	w7	Neutral	Just because the woman was wholesome-looking does not mean that she was acting in a discreet manner.	Inferential Knowledge
	w8	Contradiction	The woman was described as wholesome and wouldn't desecrate something.	Semantic
VariErr	0	Neutral	The context doesn't mention anything about desecration.	Absence of Mention
	1	Neutral	It's not clear what the woman was trying to be.	Absence of Mention
	2	Neutral	The attempt of the woman is not given in the context.	Absence of Mention
	3	Contradiction	Context is a compliment, statement is a negative comment.	Logic Conflict

Table 6: Examples of NLI items annotated in both LiveNLI and VariErr by the selected annotator groups (w1, w4, w7, w8 for LiveNLI; 0, 1, 2, 3 for VariErr).